Frohman problem — grind sheet

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Problem. Let $W = u^5v + v^6$ and $F = W \oplus W \oplus W$. Let J scale every coordinate of $\mathbb{C}^6$ by a primitive sixth root of unity. Compute

$$\langle \alpha, \beta, \gamma \rangle_{0,3}^{F,\langle J \rangle} = \int_{\overline{\mathcal{M}}_{0,3}^{F,\langle J \rangle}} \left[\overline{\mathcal{M}}_{0,3}^{F,\langle J \rangle}\right]^{\mathrm{vir}}$$

for every triple in the $2610 = 2605 + 5$ dimensional state space that passes $k + \ell + m \equiv 1 \pmod 6$. Show the numbers are not a product of three Guéré integrals of W.

Real inputs, no dummy.

$$v^{11} = v^5(5u^4 + v^6) - 5u^3(uv^5) \in (\partial W^T).$$

LTs $\{u^4, uv^5, v^{11}\}$ lock $\mu(W^T) = 11 + 15 = 26$. Locked triple: $\mu(W) = 25$, $\mu(W^T) = 26$, $|\det A| = 30$. Cubes 15625, 27000.

Selection rule $k + \ell + m \equiv 1 \pmod 6$: 36 ordered decoration triples. Slot counts are sizes, not values. OPEN cells are not 0, 1, or 6.